



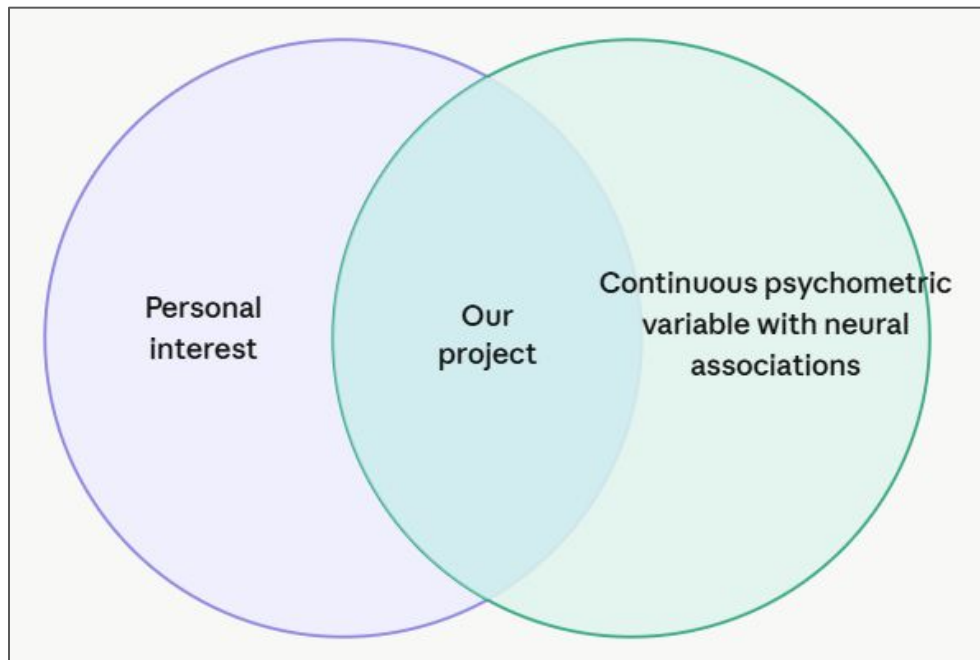
Multidimensional Approach to Examining Aggression Vulnerability: Psychosocial Factors and Structural Variations in the Amygdala and Prefrontal Cortex

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Background

- Why Aggression?



Background

- Majority of aggression research rely on **self-report measures** and **behavioural paradigms** (Roberton et al., 2012)
 - Studied in silos from biological/neural explanations for aggression
 - reactive aggression that is rapid and emotionally driven, where immediate action is prioritized over long-term consequential thinking is not able to be fully explained by self-report measures

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 - reactive aggression that is rapid and emotionally driven, where immediate action is prioritized over long-term consequential thinking is not able to be fully explained by self-report measures
- Existing neuroimaging research (Rosell & Siever, 2015; Siever, 2008) suggests that aggression is associated with:
 - Poor emotional regulation ↔ emotion-processing systems (e.g., amygdala)
 - low inhibitory control ↔ reward-processing systems
 - maladaptive decision-making ↔ executive-control systems (e.g., prefrontal cortex)
 - **BUT** relevant brain systems are often studied independently

Background

Original idea

Functional connectivity
between PFC and Amygdala



Current idea

Structural Variations in the
region of PFC and Amygdala

Background

- Smaller amygdalar volumes have been found to have higher levels of aggression and violence
 - Smaller amygdala volume has a higher correlation with violence (Johansen et al., 2011; Rosell and Siever, 2015)
 - In affective psychoses/bipolar disorder specifically, studies have detected increased volumes of amygdala (Widmayer et al., 2018) alongside aggressive behaviours

Background

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 - In affective psychoses/bipolar disorder specifically, studies have detected increased volumes of amygdala (Widmayer et al., 2018)
- Lower PFC volume (particularly orbitofrontal and dorsolateral regions) has been found to correlate with antisocial personality disorder (Raine et al., 2000; Yang & Raine, 2009)
 - increased risk of violent outcomes across personality disorder studies (Yu et al., 2012)

Research Gap

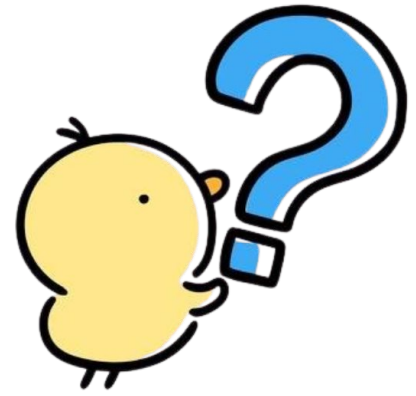


Examine the relationship between **psychosocial factors** and the **structural variations** in the relevant neural systems (**amygdala** and **prefrontal cortex**) and whether these structural characteristics are associated with **vulnerability to aggressive behaviours**.

Literature Review

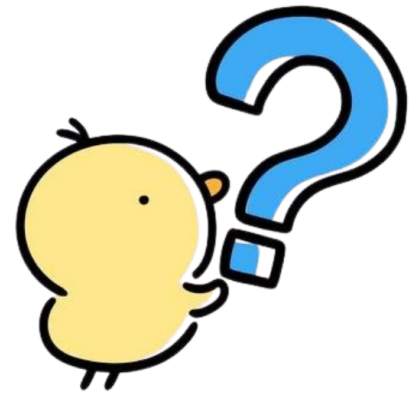
Psychosocial Factors (5)	Relationship with Aggression	Citations
Stress	↑ Stress ↑ Aggression	Anderson & Bushman, 2002
Self-efficacy	↓ Self-efficacy ↑ Aggression	Mesurado et al., 2018; Roberton et al., 2012
Social Relationships	↓ Support, ↑ Conflict, ↑ Rejection, ↑ Aggression	Heilbron & Prinstein, 2008
Executive Function	↓ Inhibition, ↑ Aggression	Choy et al., 2025; Pawliczek et al., 2013
Big 5 Personality traits	↓ Agreeableness, ↓ Conscientiousness, ↑ Aggression	Jones et al., 2011

Research Question



Is **aggression-related psychosocial factors** (stress, self-efficacy, executive control, anger/hostility, and personality) associated with **structural variation** in the **amygdala** and **prefrontal cortex**, and is this association moderated by age and gender?

Hypothesis



H1 (Main association): **Psychosocial factors** (stress, self-efficacy, executive control, anger/hostility, personality) will show associations with **structural variation** in the **amygdala** and **prefrontal cortex** (mOFC, rACC)

H2 (Moderation by age and gender): The relationship between **amygdala-PFC structural** balance and aggression-related psychosocial functioning will be **moderated by age and/or gender**

Methods

Dataset

Human Connectome Project Young Adult S1200 Release ($n = 1206$)

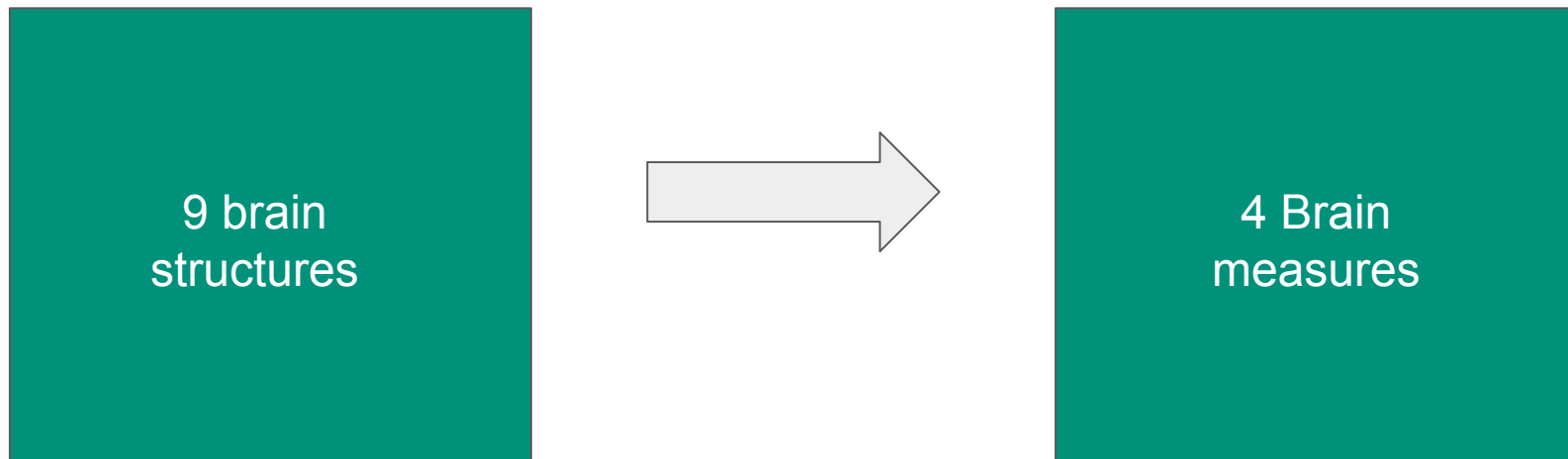
- Age: **22-35** years old
- Data collected between **2012 -2015**



HCP variables of interest

Demographics	Gender Age Range
Psychosocial factors (11)	(1) NEOFAC_A, (2) NEOFAC_C (Agreeableness, Conscientiousness) SelfEff_Unadj (Self-efficacy) PercStress_Unadj (Perceived stress) AngAffect_Unadj (Anger-Affect) AngHostil_Unadj (Anger-Hostility) AngAggr_Unadj (Anger-Physical Aggression) Flanker_Unadj (Flanker Inhibitory Control and Attention Test) Flanker_AgeAdj (Flanker Inhibitory Control and Attention Test: Age-Adjusted) PercHostil_Unadj (Perceived Hostility) PercReject_Unadj (Perceived Rejection)
Brain structures (9)	FS_L_Amygdala_Vol, FS_R_Amygdala_Vol FS_L_Medialorbitofrontal_GrayVol, FS_R_Medialorbitofrontal_GrayVol FS_L_Rostralanteriorcingulate_GrayVol, FS_R_Rostralanteriorcingulate_GrayVol FS_L_Medialorbitofrontal_Thck, FS_R_Medialorbitofrontal_Thck FS_InterCranial_Vol

HCP variables of interest



- (1) **Amygdala_Vol** = (FS_L_Amygdala_Vol + FS_R_Amygdala_Vol) / 2 → then ÷ FS_InterCranial_Vol → **Amygdala_Vol_norm**
- (2) **mOFC_GrayVol** = (FS_L_Medialorbitofrontal_GrayVol + FS_R_Medialorbitofrontal_GrayVol) / 2 → ÷ ICV → **mOFC_GrayVol_norm**
- (3) **rACC_GrayVol** = (FS_L_Rostralanteriorcingulate_GrayVol + FS_R_Rostralanteriorcingulate_GrayVol) / 2 → ÷ ICV → **rACC_GrayVol_norm**
- (4) **mOFC_Thck** = (FS_L_Medialorbitofrontal_Thck + FS_R_Medialorbitofrontal_Thck) / 2 → **mOFC_Thck** (not normalized by ICV, since cortical thickness isn't a volume measure and doesn't scale with head size the same way)

Methods

Analysis Plan

- Neuroimaging data: structural MRI
 - Compare the structural variations (volume, grey matter volume and cortical thickness) between PFC and amygdala
 - Brain ROI: Prefrontal cortex (cognitive control) and amygdala (emotion processing)
-

Methods

Data Cleaning:

Data merge & cleaning → variable normalization (bilateral averaging + ICV correction) → descriptive distribution checks & outlier detection (z-score >4 SD) → outlier removal → z-scoring

Analysis:

bivariate Pearson correlations (44 pairs) with FDR correction → composite variable construction (Aggression composite, Amygdala-PFC balance) → OLS moderation regression (Balance × Age × Gender) → visualization of interaction effect

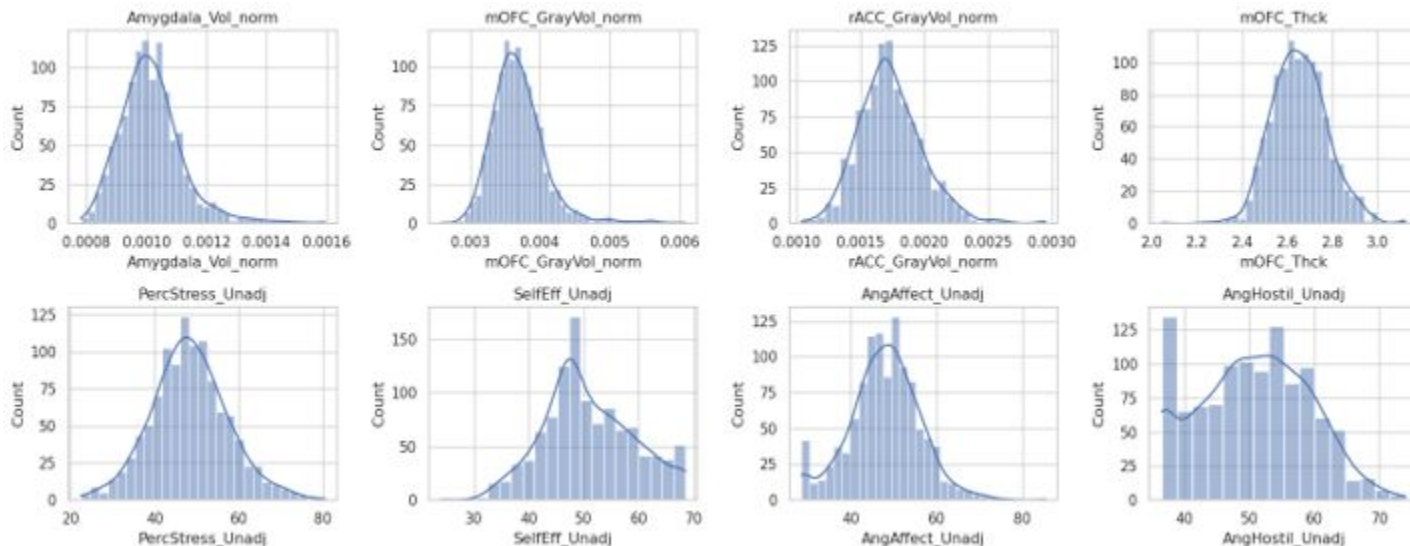
Methods

After merging FreeSurfer structural data with behavioral data and dropping missing cases ($n = 1,105$), we screened the four brain ROI measures for extreme outliers before z-scoring.

- For each of the 4 brain ROI variables, compute a z-score; flag values with $|z| > 4$ SD.
- Any subject flagged on at least one of the 4 variables was dropped from the dataset (not just that variable).
- Remaining rows were then used for z-scoring all variables and all downstream analyses.

Brain measure	Outliers ($ z > 4$)
Amygdala volume (normalized)	6
mOFC gray matter volume (normalized)	10
rACC gray matter volume (normalized)	3
mOFC cortical thickness	1

Methods

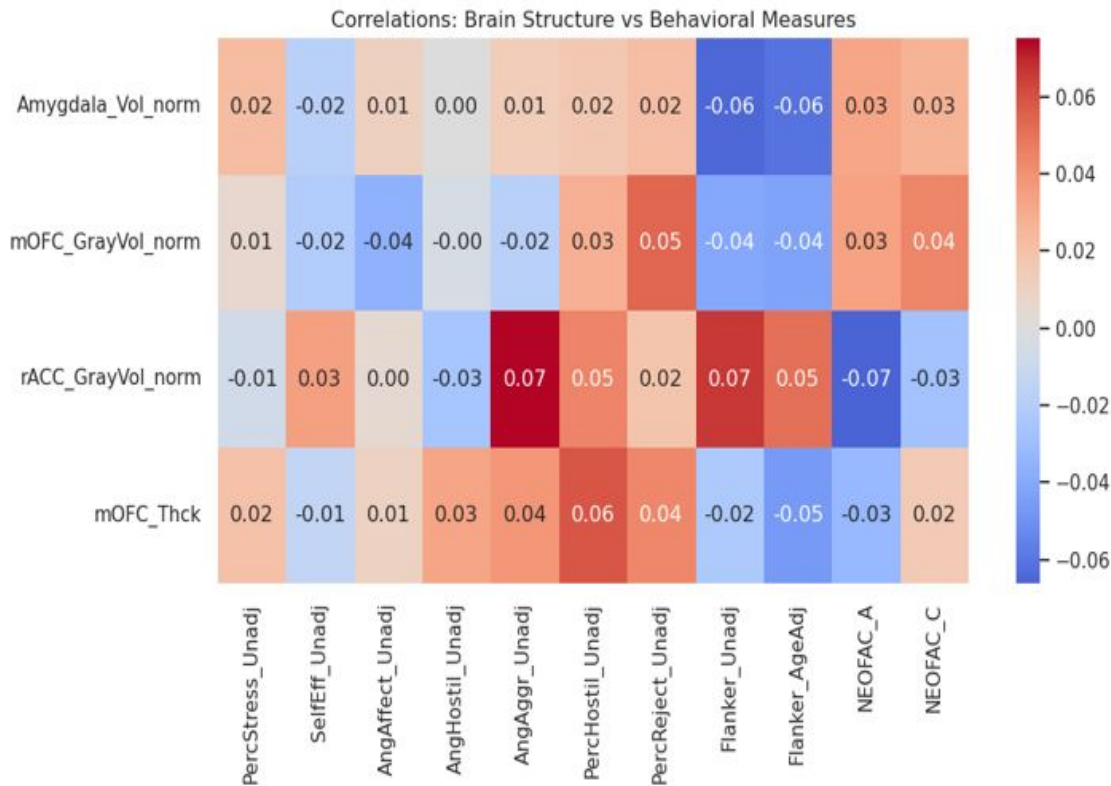


- Amygdala volume (normalized): 6 outliers ($|z| > 4$)
- mOFC gray matter volume (normalized): 10 outliers ($|z| > 4$)
- rACC gray matter volume (normalized): 3 outliers ($|z| > 4$)
- mOFC cortical thickness: 1 outlier ($|z| > 4$)

Final analytic sample after outlier removal: **n = 1,088** (17 subjects dropped across all flagged variables).

Results

Bivariate correlations ($n = 1,105$) between brain structure (amygdala, mOFC, rACC volumes/thickness, normalized for head size) and psychosocial measures were **all very weak** ($|r| \leq 0.08$).

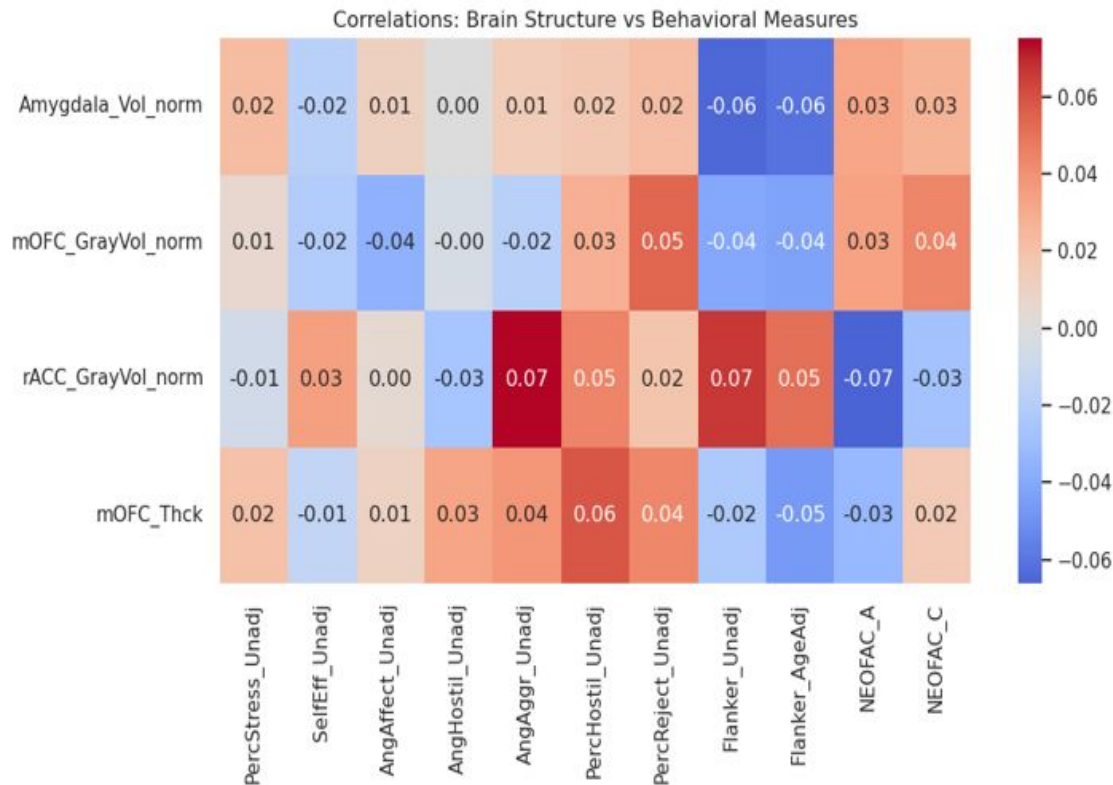


Results

After Benjamini-Hochberg FDR correction (44 comparisons):

0 of 44 pairs remained significant ($q < 0.05$).

Even the largest $r \approx 0.07$ explains under 1% of variance — a null/negligible brain-behavior pattern at the bivariate level.



Composite & Moderation Analysis

Aggression composite

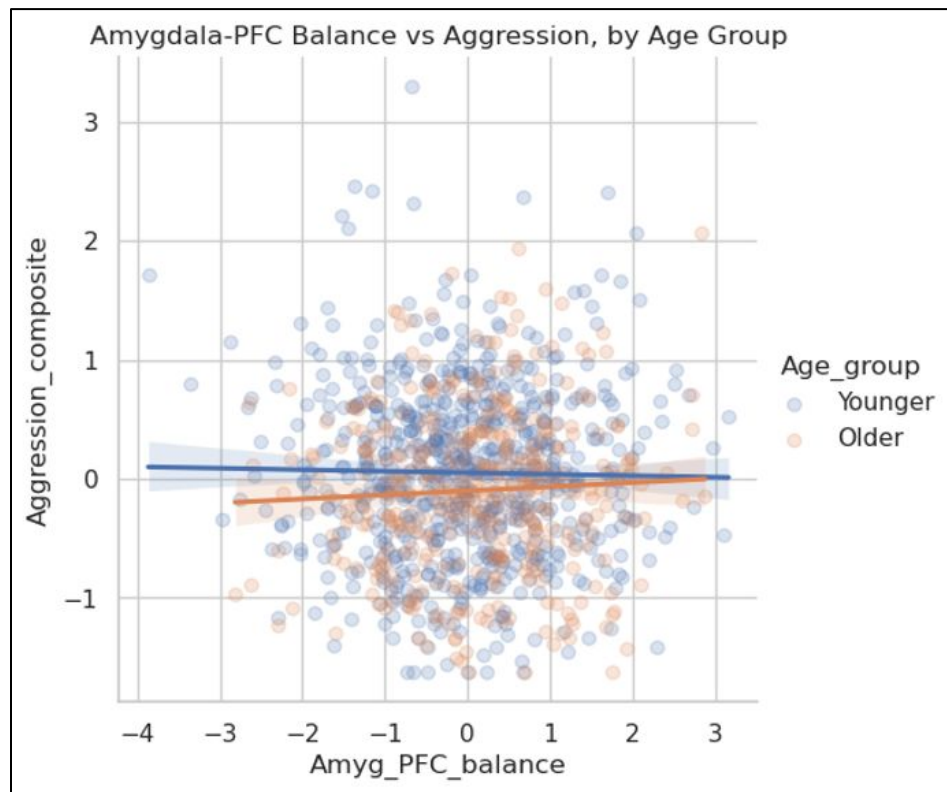
Mean of z-scored Anger-Affect, Anger-Hostility, Anger-Aggression, Perceived Hostility, and Perceived Rejection (Cronbach's $\alpha = 0.76$).

Amygdala-PFC balance

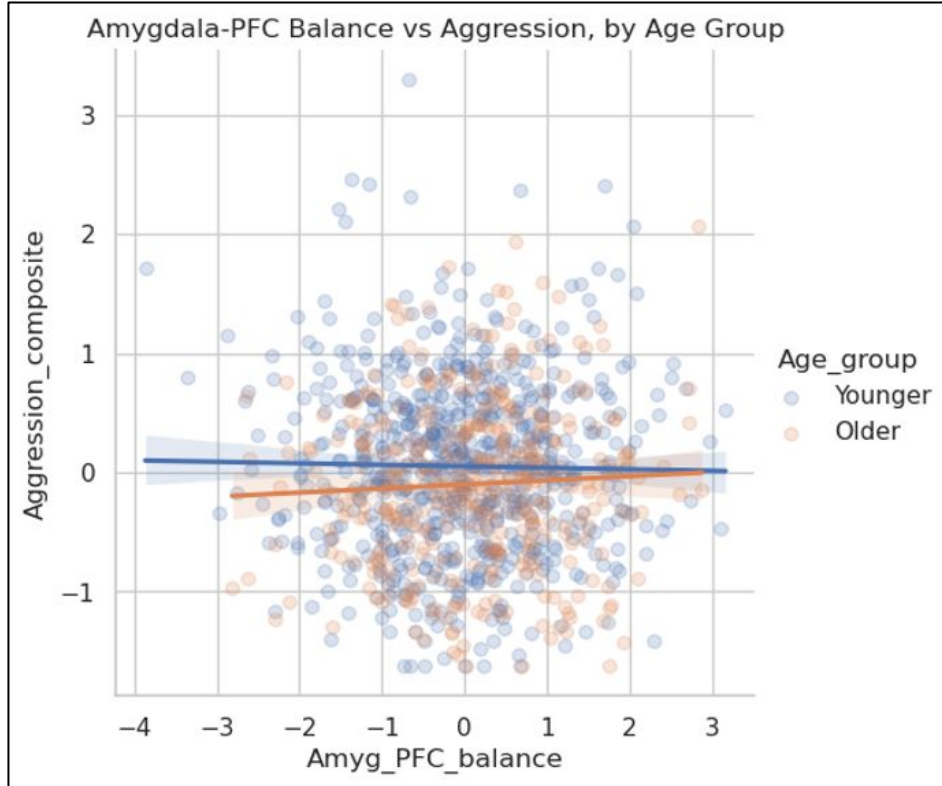
Amygdala volume (z) minus the average of mOFC and rACC gray matter volume (z). Higher = relatively larger amygdala vs. regulatory regions.

Composite & Moderation Analysis

Predictor	β	p
Amygdala-PFC balance	0.044	0.145
Gender	-0.198	< .001
Amygdala-PFC balance $\times$ Gender	-0.075	0.070
Age (z)	-0.052	0.019
Amygdala-PFC balance $\times$ Age	0.040	0.048



Composite & Moderation Analysis



Overall model: $R^2 = .033$,
 $F(5,1082) = 7.29$, $p < .0001$, $n = 1,088$.

The Amygdala-PFC balance $\times$ Age interaction is significant — the brain-behavior relationship depends on age.

Discussion

Key findings:

Simple brain-behavior associations were null

Bivariate correlations between amygdala/mOFC/rACC structure and aggression-related traits were all weak ($|r| \leq 0.08$) and none survived FDR correction (0/44).

Composite Amygdala-PFC balance score interacts with age

A model combining Amygdala-PFC balance, gender, and age significantly predicted the Aggression composite ($R^2 = .033$, $p < .0001$), with a significant Balance $\times$ Age interaction ($p = .048$) — suggesting the link between regional brain structure and aggression may emerge or shift across development rather than being a fixed, uniform effect.

Discussion

Key findings:

Structure alone is insufficient — context matters

Regional gray matter volume/thickness alone explains very little variance in aggression vulnerability; demographic moderators (age, gender) and likely functional/network-level measures carry more of the signal.

Future Directions

- Examine functional neural processes and brain network interactions, not just regional structure.
- Model age (and possibly developmental stage) as a moderator rather than a covariate.
- Test whether the Balance $\times$ Age effect replicates in samples with a wider age range.

THANK YOU

References

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Discussion

Key findings:

- Structural variations were not strong predictors
 - Structural variations in the amygdala, mOFC, and rACC showed minimal relationships with aggression-related traits
- Aggression may not be strongly reflected in regional brain structure
 - Structural measures alone may be insufficient to capture aggression vulnerability

Future research should examine integrated biopsychosocial and functional neural mechanisms (e.g., functional neural processes, brain network interactions).
